

Fundamentos matemáticos de la Causalidad

Clase 2

August 24, 2026

Tarea para hoy (aviso por campus):

Leer para el lunes 24 de agosto (i) los primeros 3 capítulos de What if?,
(ii) los primeros dos capítulos del libro de A first course in causal inference
- Peng Ding.

Quién dijo quién dijo....

we focus on the identification and estimation of causal effects in populations, i.e., numerical quantities that measure changes in the distribution of an outcome under different interventions.

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What if? Robins & Hernán

Outcome

Y

¿Quién usa cada una de las siguientes notaciones?

1. $Y(1), Y(0)$

2. $Y^{a=0}, Y^{a=1}, Y^a$

3. $Y_a, Y_{(a)}, Y^a$

¿Quién usa cada una de las siguientes notaciones?

1. $Y(1), Y(0)$ - Ding /Neal

2. Y^0, Y^1, Y^a - What if?

3. $Y_a, Y_{(a)}, Y^a$ Yo!

Cómo llamamos al tratamiento?

1. A

2. Z

3. T

Cómo llamamos al tratamiento?

1. *A* - What if?
2. *Z* - Ding
3. *T* - Yo!

Potential or conunterfacual outcomes?

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1. *Some authors prefer the term “potential outcomes” to emphasize that, depending on the treatment that is received, either of these two outcomes can be potentially*
2. *Other authors prefer the term “counterfactual outcomes” to emphasize that these outcomes represent situations that may not actually occur (that is, counter- to-the-fact situations).*

De dónde es?

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What if?

The potential outcome that is observed is sometimes referred to as a factual. Note that there are no counterfactuals or factuals until the outcome is observed. Before that, there are only potential outcomes. -

Neal

Supuestos - Assumptions

No interference

1. *Unit i 's potential outcomes do not depend on other units' treatments. This is sometimes called the no-interference assumption.*
2. *No interference means that my outcome is unaffected by anyone else's treatment. Rather, my outcome is only a function of my own treatment.*

$$Y_i(t_1, t_2, \dots, t_{i-1}, t_i, t_{i+1}, \dots, t_n) = Y_i(t_i)$$

Consistency

For each individual, one of the counterfactual outcomes—the one that corresponds to the treatment value that the individual did receive—is actually factual.

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If $A = a$, then $Y = Y^a$.

SUTVA

SUTVA You will also commonly see the stable unit-treatment value assumption (SUTVA) in the literature. SUTVA is satisfied if unit (individual) outcome is simply a function of unit treatment. Therefore, SUTVA is a combination of consistency and no interference

The Fundamental Problem of Causal Inference

It is impossible to observe all potential outcomes for a given individual.

It is fundamental because if we cannot observe both $Y_i(0)$ and $Y_i(1)$, then we cannot observe the **INDIVIDUAL** causal effect $Y_i(1) - Y_i(0)$.

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The *factual* variable is observed and the counterfactual is missing.

Causal inference and missing data

- Falta por construcción / diseño: uno de los potential outcomes en cada individuo es faltante.

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- "ausencia estructural de datos", "faltantes por diseño", "no observable por construcción"

En síntesis

Asumiendo No interferencia y consistencia, tenemos:

- Variables en el mundo causal: (Y^0, Y^1, T)
- Variables observadas: (T, Y)

Average treatment effect

$$ATE = \mathbb{E}(Y^1) - \mathbb{E}(Y^0)$$

Identificabilidad

¿Queda $ATE = \mathbb{E}(Y^1) - \mathbb{E}(Y^0)$ determinado por la distribución conjunta de (T, Y) ?

Tarea 2

La siguiente tabla representa la distribución conjunta del vector (T, Y) .

	$T = 0$	$T = 1$
$Y = 0$	0.25	0.25
$Y = 1$	0.15	0.35

$(Y^{(0)}, Y^{(1)})$	$T = 0$	$T = 1$
$(0, 0)$		
$(0, 1)$		
$(1, 0)$		
$(1, 1)$		

Exchangeability

Exchangeability is that it guarantees that the treatment groups are comparable.

$$Y^a \perp\!\!\!\perp A \quad \text{for all values } a.$$

Supuestos para identificabilidad

exchangeability means that the counterfactual outcome and the actual treatment are independent, or $Y^a \perp\!\!\!\perp A$ for all values a . Randomization is so highly valued because it is expected to produce exchangeability. When the treated and the untreated are exchangeable, we sometimes say that treatment is exogenous, and thus exogeneity is commonly used as a synonym for exchangeability.

Identificabilidad:

Completar la conjunta de (Y^0, Y^1, T) assumiendo que $Y^a \perp\!\!\!\perp T$, para $a = 0, 1$.